

Presentations, Seminar Talks, & Conferences

Diljit Singh

Research Presentations

1. Combinatorial Models of Quantum Matrices
Joint Mathematics Meetings 2018 (Session on Structure & Representations of Hopf Algebras)
2. Laplacians on Projective Fractals
Young Mathematicians Conference, 2017
3. The Intersection Number of Certain Transformation Digraphs
Macaulay Honors College Research Expo, 2017
4. Spectral Results on Projective Fractals
Cornell University, Undergraduate Research Forum, 2016
5. **Poster:** Quantum Determinant Identities via Cauchon Diagrams
University of California, Santa Barbara, Bren Hall Poster Session, 2017

Graduate Seminar Talks

6. An Introduction to Additive Number Theory **(1 Hour)**
Graduate Student Seminar, Stony Brook University, 2019
7. Atiyah's "On Analytic Surfaces with Double Points" **(1 Hour)**
Graduate Algebraic Geometry, Stony Brook University, 2019
8. Infinite Periodic Groups and Burnside Problems **(1 Hour)**
Groups & Semigroups with Applications to Theoretical Computer Science, 2018
9. Regular Local Rings are Factorial **(2 Hours)**
Algebraic Geometry Lectures, CUNY Graduate Center, 2018
10. Tverberg's Theorem (5-part seminar series) **(5 Hours total)**
Discrete Geometry Student Seminar, CUNY Graduate Center, 2018
11. Proof of the Witness Theorem **(45 Minutes)**
Topics in Computation, CUNY Graduate Center, 2017
12. Kelly's Resolution of The Sylvester–Gallai Problem of J. P. Serre **(1 Hour)**
Discrete Geometry Student Seminar, CUNY Graduate Center, 2017
13. The Golod–Shafarevich Theorem **(2 Hours)**
Number Theory Student Seminar, CUNY Graduate Center, 2017
14. On Hirzebruch's "Arrangements of Lines and Algebraic Surfaces" **(2 Hours)**
Algebraic Geometry Lectures, CUNY Graduate Center, 2017

15. Variations of the Sum–Product Problem (**2 Hours**)
Number Theory Student Seminar, CUNY Graduate Center, 2017
16. Small Multiplicative Subgroups Are Not Sumsets (**2 Hours**)
Number Theory Student Seminar, CUNY Graduate Center, 2017
17. Using Algebraic NT to Determine Rational Points on a Curve of Selmer (**2 Hours**)
Algebraic Geometry Lectures, CUNY Graduate Center, 2017
18. Infinite Descent and $E(\mathbb{Q})/2E(\mathbb{Q})$ (**2 Hours**)
Algebraic Geometry Lectures, CUNY Graduate Center, 2017
19. Proof of the Weak Mordell–Weil Theorem (**2 Hours**)
Algebraic Geometry Lectures, CUNY Graduate Center, 2017
20. Discrete Geometry Problem Session (**1 Hour**)
Discrete Geometry Student Seminar, CUNY Graduate Center, 2017
21. The Crossing Number Inequality & the Szemerédi–Trotter Theorem (**1 Hour**)
Discrete Geometry Student Seminar, CUNY Graduate Center, 2017
22. The Effective, Combinatorial, & Hilbert’s Nullstellensatz (**2 Hours**)
Algebraic Geometry Lectures, CUNY Graduate Center, 2016
23. Trivial (over $\mathbb{Q}$) Curves of Genus 1 & Failure of the Hasse Principle (**2 Hours**)
Algebraic Geometry Lectures, CUNY Graduate Center, 2016

Undergraduate and High School Aimed Talks

24. The Polynomial Method in Olympiad Problems
Stony Brook University, 2019
25. Hard Walks on Good Sequences
Fortnight Research Dinners, Brookdale, 2018
26. On (Some of) Hilbert’s 23(+1) Problems
Macaulay Splash, 2017
27. Applications of the Probabilistic and Polynomial Method to Graph Theory
Discrete Math Lecture, CUNY Hunter College, 2015
28. On Recent Results in Intersection Digraph Theory (2 Parts)
Graph Theory Lectures, CUNY Hunter College, 2015

Summer Schools & Conferences

1. **Funded** Analytic & Combinatorial Number Theory: The Legacy of Ramanujan
University of Illinois Urbana–Champaign, June 6–9, 2019
2. Combinatorial and Additive Number Theory (CANT)
CUNY Graduate Center, NYC, May 22–25, 2018
3. **Funded** Arizona Winter School 2018: Iwasawa Theory
University of Arizona, Tucson, March 3–7, 2018

4. **Funded** Joint Math Meetings 2018
San Diego, California, Jan. 10–13, 2018
5. **Funded** AMS Short Course: Discrete Differential Geometry
San Diego, California, Jan. 8–9, 2018
6. **Funded** SACNAS National Conference
Salt Lake City, Utah, Oct. 19–21, 2017
7. **Funded** ICERM Modern Math Workshop
Brown University, Salt Lake City, Oct. 18–19, 2017
8. **Funded** Young Mathematicians Conference
Ohio State University, Columbus, Aug. 11–14, 2017
9. **Funded** Thematic Program on Kähler Geometry
University of Notre Dame, June 12–16, 2017
10. **Funded** 3rd Annual Graduate Student Conference in Algebra, Geometry, & Topology
Temple University, Philadelphia, June 3–4, 2017
11. Combinatorial and Additive Number Theory (CANT)
CUNY Graduate Center, NYC, May 23–26, 2017
12. AMS Spring Eastern Sectional Meeting
Hunter College, NYC, May 6–7, 2017
13. Mathfest
CUNY Graduate Center, NYC, March 10, 2017
14. **Funded** Math Day
Stony Brook University, Long Island, Nov. 12, 2016
15. Combinatorial and Additive Number Theory (CANT)
CUNY Graduate Center, NYC, May 24–27, 2016
16. Manhattan Algebra Day
CUNY Graduate Center, NYC, Dec. 9, 2015
17. International Conference on Harmonic Analysis and Applications (ICHAA)
CUNY Graduate Center, NYC, June 1–5, 2015
18. Combinatorial and Additive Number Theory (CANT)
CUNY Graduate Center, NYC, May 19–22, 2015
19. Mathfest
CUNY Graduate Center, NYC, Nov. 11, 2014